

Canonical two-chunk regional simulations (en)

Canonical two-chunk regional simulations — ULVZ project guide

Read this page first. This project-local guide covers only the accepted canonical mode: NCHUNKS=2, two adjacent 90°×90° faces, AB central, AC joined on the supported local side, a single center/gamma orientation, and the accepted patch. It is not an upstream SPECFEM3D_GLOBE promise. canonical_90deg_fixture_ready=true; general_two_chunk_mode_classification=B.

Do not extrapolate. Non-90° or unequal widths, other attachment sides, nonadjacent faces, independent orientations, arbitrary two-chunk topologies, and three-/general multi-chunk modes are unverified. The planner is read-only: it does not replace mesher, Stacey, or waveform acceptance. Kim/Song exact reconstruction still requires author inputs.

1. What this mode is, and the words used below

A regional simulation computes only selected cubed-sphere faces. One chunk is the usual regional case; six chunks form the full sphere. This two-chunk mode is useful only when the source, receivers, and target paths do not fit one 90° face but do fit these two adjacent faces. It is not a free-form mesh.

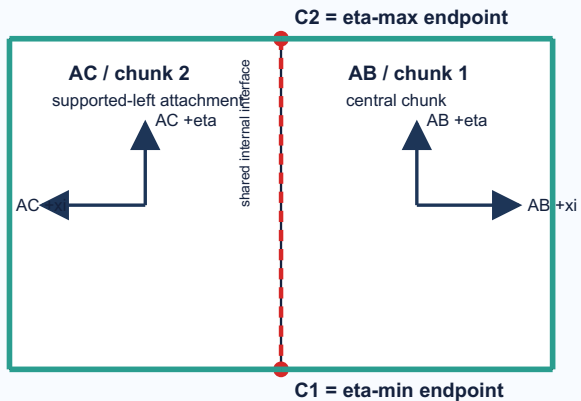
Term	Plain meaning and location	Why a user must care / visible failure
AB / AC	The two faces used here. AB is central chunk 1; AC is attached chunk 2.	They identify the owner of a source or receiver. Swapping them changes supported topology.
AB-AC interface	The common interior face: AB xi-min meets AC xi-max.	It exchanges wavefield data; it is not an exterior wall and must never absorb.
C1 / C2	The two interface ends: C1 is eta-min; C2 is eta-max. Eta is face-local, not geographic south/north.	Endpoint errors cause missing or wrongly connected corner data and may vary by MPI layout.
MPI rank / rank zero	A rank is one MPI process; rank 0 is a valid process.	Rank 0 must not be treated as an unused slot.
buffer / two-member path	A buffer carries endpoint data between the two ranks meeting at C1 or C2.	Each endpoint needs exactly two reciprocal members in this mode.
three-member / BC path	Historical general-face corner logic included a BC participant not in this two-face domain.	It can create a false connection or conceal a missing endpoint.
Stacey face / sponge	Stacey absorbs on an exposed mesh face. A global sponge is a high-attenuation region in a six-chunk global model.	Neither may be put on AB-AC.

Implementation background. Before the accepted patch, C1 had no eta-min two-member record and C2 retained a BC/rank-zero three-member path. The patch creates two endpoint records, uses INVALID_RANK only for unused slots, and segments the xi-constant interface with NPROC_ETA.

2. Geometry: local attachment is not a map direction

Canonical 90° two-chunk geometry: local topology is not a geographic map

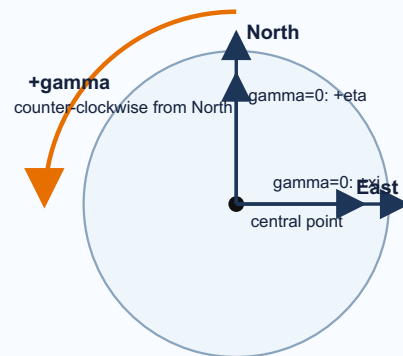
A. Unrotated local topology (attachment definition)



Dashed red: MPI field exchange only — never sponge or Stacey.
Green solid faces: exposed external faces eligible for regional Stacey roles.

B. Whole-system orientation (tangent-plane view)

View from outside Earth looking down at the central point.



gamma 0°: +eta North, +xi East
gamma 90°: +eta West, +xi North
gamma 180°: +eta South, +xi West

Source basis: current Euler matrix (src/shared/euler_angles.f90), regional manual §5, accepted canonical fixture. AB/AC names denote cubed-sphere face roles, not compass directions.

The left panel is a **local topology drawing**, not a map. “Supported-left” means that AC connects to AB’s designated local edge: AB xi-min joins AC xi-max. Once center latitude/longitude and gamma rotate the complete system, AC may appear west, east, north, or south on a map. Say “attached to the specified local edge”, not “geographically left”.

The dashed AB–AC line is internal. C1 is eta-min and C2 eta-max. Remaining lateral faces are exposed. An endpoint can share a *node* with an external Stacey face without making the internal face absorbing: face roles are different. Accepted topology evidence found zero internal AB–AC Stacey faces at shallow, mid-mantle, and CMB-near depths.

Current source stops when NCHUNKS > 1 and ANGULAR_WIDTH_XI_IN_DEGREES is not 90. It permits a non-90 eta width only for exactly two chunks, but this project has not validated that case: set both widths to 90.

3. Center coordinates and GAMMA_ROTATION_AZIMUTH

CENTER_LATITUDE_IN_DEGREES and CENTER_LONGITUDE_IN_DEGREES place AB's center. GAMMA_ROTATION_AZIMUTH rotates the **complete AB+AC system**; AC has no independent orientation. The official regional manual says gamma is measured counter-clockwise from due north. Current euler_angles.f90 and the project planner use the same sign.

Viewing convention: look from outside the Earth toward the selected center, with north up and east right in the tangent plane. Positive gamma is counter-clockwise. With ELLIPTICITY=false, and center=(0°,0°):

Gamma	+eta	+xi	Meaning
0°	north	east	reference orientation
90°	west	north	complete system rotated counter-clockwise
180°	south	west	half turn

With ellipticity enabled the source first converts geographic latitude to geocentric colatitude; this does not change gamma's sign. Audit a source and all stations after every center/gamma change. Never rotate AC separately.

Determine AC's actual geographic position after setting parameters

There is no AC latitude, longitude, or gamma parameter. For canonical geometry, the four inputs NCHUNKS=2, both 90° widths, AB center latitude/longitude, and gamma determine every AB and AC boundary point uniquely: SPECFEM first defines the fixed local AB--AC pair, then applies the same Euler transform to the whole pair. AC is therefore not a second region that the user places.

For a selected center/lon/gamma, use two_chunk_planner plan with the same minimum and maximum for each search range. Its map.png shows the transformed AB/AC outline and interface; geometry_audit.json and candidates.json report source/station chunk labels, local xi/eta, interface, C1/C2, and exposed-boundary margins. This is the operational way to decide whether the intended geographic region falls in AB, AC, the shared interface, or outside. It is a pre-mesh geometric check, not a replacement for later mesh/Stacey/waveform validation.

At a geographic pole, longitude and gamma are not a unique parameterization. The planner deliberately normalizes a polar center to longitude 0° and adjusts gamma to preserve the same physical AB+AC orientation. Compare the plotted outline, not only the three printed numbers, when reviewing a polar case.

4. Install the accepted patch safely

The package is patches/specfem3d_globe/two_chunk_endpoints/. Its JSON manifest is the hash authority. It was validated against nested revision 9c312cb2c991b47484a7f302775f4f01ed9470f8. Baseline target SHA-256 is fd4137713e55e14ec664a9d55487b64c2b9bf73499c1f82780f1f5a6e63b088f; applied target SHA-256 must be 8c64f1d1d415ec6c0792f06474dafcfc698da6ee03ecd21bfd4fdc90b64857.

Minimum installation

Run from the project root. Stop on a hash or context mismatch.

```
PATCH="$(pwd)/patches/specfem3d_globe/two_chunk_endpoints/specfem3d_globe_two_chunk_endpoints.patch"
patches/specfem3d_globe/two_chunk_endpoints/verify_patch.sh specfem3d_globe
git -C specfem3d_globe apply --check $PATCH
git -C specfem3d_globe apply $PATCH
sha256sum specfem3d_globe/src/meshfem3D/create_chunk_buffers.f90
git -C specfem3d_globe apply --reverse --check $PATCH
```

The verifier must report baseline hash, single-file scope, no DEBUG change, and a clean apply check. If the target already has the candidate hash, the reverse dry-run identifies it as applied; do not apply twice. To reverse intentionally: run git -C specfem3d_globe apply --reverse \$PATCH, then rebuild from clean objects.

Strict traceability and manual replacement

Record nested revision, baseline hash, patch SHA-256 4496cea542d26f38575ec1fa9ae28635ec2a201958eb898702331c7db5fe4a60, forward check, applied hash, and reverse check. Prefer git apply because it checks context and leaves an auditable diff. Do **not** normally replace the complete create_chunk_buffers.f90 file. If site policy requires it, preserve the original, verify the exact baseline hash, compare the patch diff, verify the candidate hash, and retain a tested restore path. Copying a file is not two-chunk runtime acceptance. This patch does not cleanly apply to v8.0.0.

5. Inputs, external absorption, source, and stations

Start from current specfem3d_globe/DATA/Par_file. The teaching copy is cases/two_chunk_canonical_90deg/DATA/Par_file.

Setting	Canonical two-chunk instruction
NCHUNKS	Set 2. Total ranks = 2NPROC_XNPROC_ETA. Accepted fixtures: 2, 8, 12 ranks.
ANGULAR_WIDTH_XI_IN_DEGREES, ANGULAR_WIDTH_ETA_IN_DEGREES	Set both 90.d0.
CENTER_LATITUDE_IN_DEGREES, CENTER_LONGITUDE_IN_DEGREES, GAMMA_ROTATION_AZIMUTH	Place/rotate whole domain; use a fixed-geometry planner check after every change.
NEX_XI, NEX_ETA, NPROC_XI, NPROC_ETA	NEX=96 and 2×2/chunk are the accepted teaching fixture, not generic cheap production accuracy. Keep source divisibility constraints.
MODEL, OCEANS, ELLIPTICITY, TOPOGRAPHY, GRAVITY, ROTATION, ATTENUATION	Choose physical assumptions deliberately. Teaching switches are not a scientific prescription.
RECORD_LENGTH_IN_MINUTES, NTSTEP_BETWEEN_OUTPUT_SEISMOS, NTSTEP_BETWEEN_OUTPUT_SAMPLE	Select duration/cadence from a target signal and documented external-return assessment.
ABSORBING_CONDITIONS	Set .true. for canonical two chunks. Mesher builds Stacey arrays only for exposed faces.
ABSORB_USING_GLOBAL_SPONGE, SPONGE_LATITUDE_IN_DEGREES, SPONGE_LONGITUDE_IN_DEGREES, SPONGE_RADIUS_IN_DEGREES	Set sponge flag .false. Current source stops if it is true with NCHUNKS=2.
REGIONAL_MESH_CUTOFF, REGIONAL_MESH_CUTOFF_DEPTH, REGIONAL_MESH_ADD_2ND_DOUBLING	Optional radial controls; retain allowed current-template values and recheck mesh/Jacobians.
LOCAL_PATH, LOCAL_TMP_PATH, SAVE_MESH_FILES, OUTPUT_SEISMOS_*	Plan new-run database, diagnostic, and waveform storage.

What actually absorbs in two chunks

Global sponge is a six-chunk global-model option. read_parameter_file.F90 reads SPONGE_* only when its flag is true, while read_compute_parameters.f90 stops any non-six-chunk request with “Please set NCHUNKS to 6 in Par_file to use ABSORB_USING_GLOBAL_SPONGE” . Its Qmu modification is reached through the ATTENUATION model path; it is not a local side-face condition.

For NCHUNKS=2, create_regions_mesh.F90 invokes Stacey setup when ABSORBING_CONDITIONS=.true. get_absorb.f90 includes AC xi-min, AB xi-max, and exposed eta faces; it excludes AB xi-min and AC xi-max, the shared interface. A radial bottom face is added only where the source logic calls for it (outer core or regional cut-off). Therefore **AB–AC is never sponge, Stacey, or any absorbing face**. If a result says otherwise, stop for a face-role audit. The exact parameter guards and call path are documented in [the absorbing-boundary source audit](#).

CMTSOLUTION contains PDE, event name, time shift, half duration, latitude, longitude, depth, and Mrr Mtt Mpp Mrt Mrp Mtp. STATIONS records are station network latitude longitude elevation burial. Sources/stations may be in either face, but avoid exact interface, endpoint, element face/edge, or exterior-face placement. Do not prescribe a fixed angular safety distance: estimate target arrival and earliest external return.

6. Custom simulation workflow and teaching fixture

Do not mix these two workflows. The first is for **your own prepared input**; the second only checks or runs the project's fixed teaching case. Neither is formal patch acceptance. Every writing command uses a new output directory.

A. Plan and run your own canonical configuration

1. Check source and patch (section 4). Prepare your own DATA/Par_file, CMTSOLUTION, and STATIONS; set two 90° widths, ABSORBING_CONDITIONS=.true., and ABSORB_USING_GLOBAL_SPONGE=.false..
2. If geometry is not yet chosen, use the package planner guide for a bounded center/lon/gamma search. Once a candidate is chosen, use the fixed-range command below to display and classify that exact geometry. Replace all values marked for replacement; the output directory must not exist.

```
export ULVZ_PYTHON=/import/freenas-m-01-seismology/xjiang/software/anaconda3/envs/ulvz-specfem/bin/python
SPECFEM_ROOT=$(pwd)/specfem3d_globe"
INPUT_DATA="/absolute/path/to/YOUR_CASE/DATA"
CENTER_LAT=20.0
CENTER_LON=160.0
GAMMA=30.0
ANALYSIS_END_S=1900
PLAN_DIR="results/two_chunk_geometry_$(date -u +%Y%m%dT%H%M%S)"
PYTHONPATH=packages/two_chunk_planner/src $ULVZ_PYTHON -m two_chunk_planner plan \
--cmtsolution "$INPUT_DATA/CMTSOLUTION" --stations "$INPUT_DATA/STATIONS" \
--par-file "$INPUT_DATA/Par_file" --target-energy-end-s "$ANALYSIS_END_S" \
--latitude-range "$CENTER_LAT,$CENTER_LAT" --longitude-range "$CENTER_LON,$CENTER_LON" \
--gamma-range "$GAMMA,$GAMMA" --output "$PLAN_DIR"
```

3. Open PLAN_DIR/map.png to see the actual geographic AB/AC boundaries. Read the chosen candidate's point classifications before continuing. If no feasible candidate is returned, or a required point is outside/on an endpoint/external boundary, stop and change geometry. Review—not blindly copy—recommended_Par_file.inc, then transfer its center/lon/gamma and any deliberately selected mesh settings into your own Par_file.
4. Run your own input only after the preceding review. NPROCS=8 is the validated 2×2-per-chunk example; change it only to a compatible layout.

```
NPROCS=8
SPECFEM_ROOT=$(pwd)/specfem3d_globe"
INPUT_DATA="/absolute/path/to/YOUR_CASE/DATA"
RUN_DIR="results/my_two_chunk_run_$(date -u +%Y%m%dT%H%M%S)"
test ! -e "$RUN_DIR" || { echo "refusing to overwrite $RUN_DIR" >&2; exit 2; }
mkdir -p "$RUN_DIR/DATA" "$RUN_DIR/DATABASES_MPI" "$RUN_DIR/OUTPUT_FILES"
cp -R "$INPUT_DATA/." "$RUN_DIR/DATA/"
(cd "$RUN_DIR" && mpirun -np "$NPROCS" "$SPECFEM_ROOT/bin/xmeshfem3D")
(cd "$RUN_DIR" && mpirun -np "$NPROCS" "$SPECFEM_ROOT/bin/xspecfem3D")
```

xmeshfem3D produces the mesh and DATABASES_MPI; xspecfem3D reads those databases. This source tree has no separate xgenerate_databases command.

B. Check or run the fixed teaching fixture

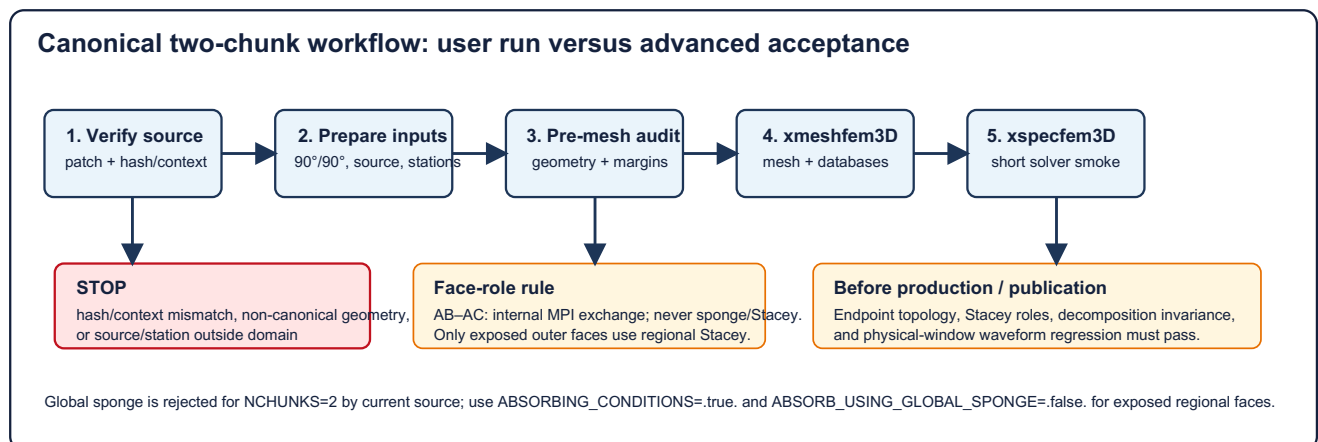
audit_geometry.py is intentionally restricted to cases/two_chunk_canonical_90deg/DATA (center 90°, longitude 90°, gamma 0°). run_canonical.sh copies that same teaching DATA; it is not a reviewer and must not be used to launch your own inputs. Its dry-run is only a command preview for the teaching fixture.

```
SPECFEM_ROOT=$(pwd)/specfem3d_globe"
$ULVZ_PYTHON cases/two_chunk_canonical_90deg/audit_geometry.py --validate-only --specfem-root "$SPECFEM_ROOT"
bash cases/two_chunk_canonical_90deg/run_canonical.sh --specfem-root "$SPECFEM_ROOT" \
--run-dir "results/two_chunk_canonical_run_$(date -u +%Y%m%dT%H%M%S)" --dry-run
```

Expected: the teaching audit reports status pass and total MPI ranks 8; its dry-run prints the 8-rank xmeshfem3D then xspecfem3D commands without creating the run directory. Remove --dry-run only to reproduce that teaching fixture, not to run a custom study.

7. Full production or release acceptance

Keep these checks separate from first use.



The diagram's pre-mesh step is the custom-input planner workflow in section 6A. The teaching audit and teaching runner in section 6B are separate fixture tools.

Level	Required evidence	Stop condition
Must	Patch context; canonical widths; in-domain points; finite mesher/solver output	mismatch, non-90° geometry, outside point, or mesher failure
Changed patch/source	C1/C2 reciprocal two-member paths; no INVALID_RANK member; zero internal AB-AC Stacey faces	missing endpoint, BC/three-member path, or internal absorbing face
Decomposition claim	Equivalent 2/8/12 physical fingerprints, connectivity, materials, buffers, face roles	unexplained rank difference
Waveform claim	Valid fixed physical window; no shift; no NaN/Inf; NRMS and relative energy $\leq 1e-5$	endpoint anomaly, invalid window, arrival shift, threshold failure
Recommended science	Target-arrival and external-return assessment; margins/path coverage	boundary contamination risk
Developer evidence	One-/six-chunk regression and controlled reversed reproduction	no broad causal claim without them

Accepted v3 achieved maximum NRMS 2.92e-6 and relative energy difference 2.86e-6 at 2/8/12 ranks. The earlier v2 [0,25] s window was before an approximately 52 s first arrival; it was invalid, not a reason to loosen gates.

8. Resolution and MPI choices

NEX_XI/NEX_ETA are lateral spectral-element counts; GLL points lie inside elements. Radial layers/doubling, model complexity, source bandwidth, timestep, and output cadence also matter. NEX=96 is an accepted canonical geometry fixture, not automatically a low-cost smoke mesh or production prescription.

Per chunk, decomposition is $NPROC_XI \times NPROC_ETA$; total ranks are $2NPROC_XINPROC_ETA$. Accepted fixtures are $1 \times 1/\text{chunk}=2$, $2 \times 2/\text{chunk}=8$, and $2 \times 3/\text{chunk}=12$. Other compatible layouts are not project-validated. If mesher stops, choose compatible NEX/ranks; never bypass the source checks.

9. What each audit actually checks

Tool or record	Input	Output / pass meaning	It does not prove
verify_patch.sh	worktree and patch	source SHA, one-file scope, no DEBUG, clean apply context	topology or waveform
audit_geometry.py	fixed teaching DATA/ only	static teaching-fixture input/hash/membership check	arbitrary user center/gamma, databases, Jacobians, Stacey, waveforms
two_chunk_planner plan	user source/stations, candidate geometry, optional path/window	deterministic center/gamma candidates; actual AB/AC map and point classification	mesher, solver, or production-safe return time
stacey.bin analysis	generated databases	face roles; internal AB-AC count must be zero	science window
acceptance reports	controlled diagnostic runs	ownership, reciprocity, fingerprints, fixed-window metrics	general two-chunk support

There is no single “run audit tool” . Use the chain: source integrity → geometry → mesh/database face roles → solver/waveforms.

10. One-page checklist

- ☐ Patch hash/context and git apply --check agree.
- ☐ NCHUNKS=2; both widths 90; AB central and AC on specified local edge.
- ☐ ABSORBING_CONDITIONS=.true.; ABSORB_USING_GLOBAL_SPONGE=.false.
- ☐ Total ranks = $2NPROC_XINPROC_ETA$; NEX compatible.
- ☐ Source/stations are in-domain and not exactly on interface/C1/C2/faces.
- ☐ Window contains target arrival and documented return risk; it is not a universal window.
- ☐ Mesher completes with intended dimensions and acceptable Jacobians.
- ☐ New DATABASES_MPI/stacey.bin exist; required face-role audit finds zero internal AB-AC Stacey faces.
- ☐ Solver traces are finite and physical.
- ☐ Any symmetry claim has fingerprints and unshifted fixed-window metrics.

11. Troubleshooting

Symptom	Likely cause	Check	Safe response
“Please set NCHUNKS to 6 ... GLOBAL_SPONGE”	sponge enabled in two chunks	ABSORB_USING_GLOBAL_SPONGE	set .false.; use external Stacey
Claimed Stacey interface	node/face confusion or defect	face-role record	require zero AB-AC face records; stop if nonzero
Patch verifier fails	unmatched or already-applied source	baseline/candidate SHA, reverse check	do not force/overwrite; revalidate version
Mesher rejects ranks/NEX	incompatible decomposition	Par_file and stop text	choose compatible layout
Point outside	center/gamma or coordinates wrong	planner for custom input; teaching audit for fixture	move point or complete system, then rerun the applicable check
Zero/invalid trace	excluded receiver/input/solver failure	receiver list/log	correct inputs in a new run
Poor pre-arrival comparison	invalid window	travel-time assessment	redesign window; never shift traces
Rank disagreement	settings/mesh defect	fingerprint/raw traces	reproduce canonical inputs and stop if unexplained

12. Teaching case, glossary, and evidence

cases/two_chunk_canonical_90deg/ is a teaching fixture, not Kim/Song Hawai’i input or a production model. It contains NEX=96, 2×2 /chunk, an isotropic 50 km source, stations in both faces, audit_geometry.py, figure source, and the deliberate runner. Static audit should report seven in-domain points and eight total ranks. Its audit and runner do not audit or launch a custom input case.

Glossary. An external boundary is exposed and may be Stacey. The internal interface exchanges wavefield and can never be sponge/Stacey. C1/C2 are interface endpoints; shared nodes do not merge face roles. A fingerprint is a physical mesh summary, not a count of MPI files.

Evidence. Official regional behavior is in specfem3d_globe/doc/USER_MANUAL/05_regional_simulations.tex. Current geometry is in euler_angles.f90, write_profile.f90, and read_compute_parameters.f90. Absorption is in read_parameter_file.F90, create_regions_mesh.F90, get_absorb.f90, get_model.F90, and meshfem3D_models.F90. Patch hashes are in patches/specfem3d_globe/two_chunk_endpoints/. Accepted topology and waveform closure are in docs/two_chunk_corner_topology_acceptance.md and docs/two_chunk_waveform_symmetry_closure.md. The documentation-change record is [the revision report](#).